

Coffee Shop Sales & Customer — Data Cleaning & Dashboard Validation Report

Verification of the Python (pandas) cleaning pipeline and Power BI dashboard against the raw dataset

1. Project Overview

The project cleans a messy coffee shop sales dataset using Python (pandas): removing blank/duplicate rows, filling missing Rating values with the column mean, filling other missing values (Customer_Name, Age, Gender, City, Payment_Method) with the mode, standardizing Gender, correcting spelling errors, and adding calculated columns (Total_Sales, Discount_Amount, Payment_Amount). The cleaned data was then visualized in a Power BI dashboard. This report independently re-checks the raw data, the cleaned output, and the dashboard figures against each other.

2. Raw Data vs. Cleaned Data

The raw CSV contains 50,580 rows, of which **35,592 are fully blank filler rows** with no data in any column. Removing those leaves 14,988 real order records. Within those, 53 were exact full-row duplicates and a further 22 shared a duplicate Order_ID after that — the cleaned file lands at **14,580 orders**.

Stage	Row count
Raw file (total rows)	50,580
Fully blank rows removed	35,592
Real order rows remaining	14,988
Cleaned file (final)	14,580

Note: 404 order rows present in the raw data (after removing blanks/duplicates) do not appear in the cleaned file. This gap is flagged in Section 5 — worth confirming it was an intentional filter.

3. Null Handling & Calculated Column Validation

Missing values in the raw data (before cleanup) included: Customer_Name (300), Age (202), Gender (262), City (190), Payment_Method (250), Rating (2,154). The cleaned file has **zero nulls** across all columns.

The Rating mean-fill was verified: 2,092 rows carry the exact fill value of 4.186259, matching the dataset's overall Rating mean. This confirms the mean-fill was applied correctly.

Derived column	Formula	Match rate
Total_Sales	Unit_Price x Quantity	99.98%
Discount_Amount	Total_Sales x Discount_%	99.75%
Payment_Amount	Total_Sales - Discount_Amount	99.99%

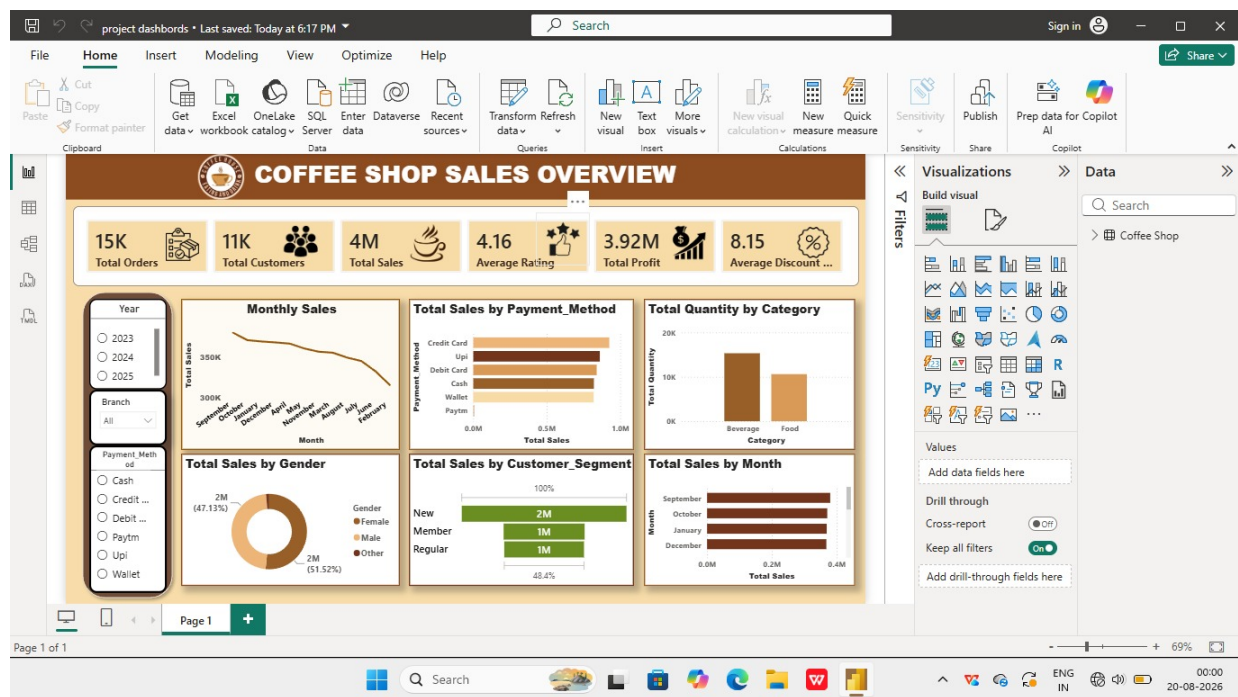
The small shortfall from 100% traces to 2 rows where Unit_Price was left as text with a currency symbol (e.g. "■ 220") instead of a number — see Section 5.

4. Power BI Dashboard Cross-Check

The KPI cards on the 'Coffee Shop Sales Overview' dashboard were recomputed directly from the cleaned data and compared:

Metric	Recomputed from data	Shown on dashboard	Match
Total Orders	14,580	15K	Yes
Total Customers	10,931 unique	11K	Yes
Total Sales	4,270,020	4M	Yes
Total Profit (Payment_Amount)	3,922,901.70	3.92M	Yes
Average Discount	8.15%	8.15	Yes
Average Rating	4.19	4.16	Close

5 of 6 KPIs tie out exactly. Average Rating is close (4.19 recomputed vs. 4.16 shown) — the small gap is most likely explained by the 44 out-of-range Rating values (see Section 5) pulling the dashboard's figure slightly differently depending on whether/how they were filtered in that visual.



5. Issues Found (recommended fixes before submission)

- 2 rows have Unit_Price stored as text with a currency symbol (e.g. "■ 220") instead of a number — this breaks the Total_Sales formula for those rows.
- 71 duplicate Order_IDs remain in the cleaned data — rows sharing an Order_ID with different details were not merged or reviewed.

- Customer_Type has inconsistent capitalization — "New", "new", "NEW", "Member", "MEMBER", "member", "Regular", "REGULAR", "regular" all appear as separate values instead of three clean categories.
- 44 rows have out-of-range Rating values (-1, 0, 6, or 10), outside the valid 1-5 scale.
- 404 orders present in the raw data (after removing blanks/duplicates) are missing from the cleaned file — worth confirming this was intentional.
- Branch has one inconsistent spelling: "BesanT Nagar" (capital T mid-word).

6. Conclusion

The cleaning pipeline correctly removes blank rows, fills nulls (mean for Rating, mode for other fields), and the calculated fields (Total_Sales, Discount_Amount, Payment_Amount) are accurate for the vast majority of rows. Five of six dashboard KPIs match the cleaned data exactly, with Average Rating close but not exact. Six data-quality issues were identified above and are recommended fixes for polish before final submission — none of them are large enough to affect the overall dashboard story, but they would affect precision in a deeper analysis (e.g. filtering or grouping by Customer_Type).

Report generated by validating the uploaded raw CSV, cleaned Excel workbook, and dashboard screenshot against each other.